

Elementary Problems Graph Theory Solutions

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Basic Work

1. When we compute the sum $d_1 + d_2 + \cdots + d_n$, each handshaking is counted twice since there are two people involved in a handshaking. So the total number of handshakings is $\frac{d_1 + d_2 + \cdots + d_n}{2}$.
2. The answer is $n - 2$. WLOG assume A and B do not shake hands with each other. Then at least 2 people do not shake hands with everyone else. The maximum case can be attained when every pair of people except A and B shake hands with each other.
3. The number of friends that each person has can only be $0, 1, \dots, n - 1$. Suppose on the contrary that no two people at the meeting have the same number of friends. Then the number of friends of the n people run through all possibilities from 0 to $n - 1$ since there are only n distinct possibilities. In particular, there is a person which has no friend, and there is a person who is a friend to everyone. This is clearly a contradiction. Thus, there must be two of them having the same number of friends.
4. Using terminologies in graph theory, we need to prove the existence of a cycle given that the number of edges is at least the number of vertices. We prove this by induction on n .

It is not possible to have $n = 1, 2$. When $n = 3$, the 3 vertices must form a cycle. Assume the assertion holds for some n . Suppose there are $n + 1$ vertices and at least $n + 1$ edges. If there exists a vertex V with degree ≤ 1 , then we remove V together with the edge (if exists) incident to V . This leaves n vertices with at least n edges. By the inductive hypothesis, we can find a cycle among these n vertices, which is a cycle of the original graph.

Suppose the degree of each vertex is at least 2. Then we start from an arbitrary vertex V_1 and find an adjacent vertex V_2 . Since the degree of V_2 is ≥ 2 , we can find another adjacent vertex V_3 . We repeat this process to find a vertex V_{j+1} adjacent to V_j . Since there are finitely many vertices, we must end up with a vertex V_k such that $V_{k+1} = V_j$ for some $j = 1, 2, \dots, k - 1$. In that case, we find a cycle $V_j - V_{j+1} - \cdots - V_k - V_j$. So the assertion holds by induction.

5. We use terminologies in graph theory. Since the total degree is $2n$, the number of edges is n . This shows there is a cycle in the graph. All vertices in this cycle are only adjacent to the 2 neighbouring vertices in the cycle. Thus, if we remove this cycle, then each of the remaining vertices (if exist) still has degree 2. Repeating the process, we can form more and more disjoint cycles until all vertices have been used. Thus, the graph is a union of some disjoint cycles.

6. Using terminologies in graph theory, we have a simple graph with n vertices such that each vertex has degree $\geq \frac{n-1}{2}$. We need to prove that the graph is connected.

Suppose on the contrary that the graph is disconnected. Then there are at least 2 components. By the pigeonhole principle, there exists a component with at most $\frac{n}{2}$ vertices. Each vertex in this component can be adjacent to at most $\frac{n}{2} - 1$ other vertices. This contradicts the condition that its degree is at least $\frac{n-1}{2}$. So the graph is connected.

7. Using terminologies in graph theory, we have a simple connected graph with an edge AB . We need to prove that AB is a bridge if and only if there is no cycle including the edge AB .

Firstly, suppose there is a cycle $A - B - V_1 - V_2 - \dots - V_k - A$. If we remove the edge AB , then we can still travel from A to B through the edges $A - V_k - V_{k-1} - \dots - V_1 - B$. So AB is not a bridge.

Secondly, suppose AB is not a bridge. After the edge AB is removed, there is still a path $A - V_k - V_{k-1} - \dots - V_1 - B$. We may assume each edge is used at most once. Then the original graph consists of the cycle $A - V_k - V_{k-1} - \dots - V_1 - B - A$.

So the equivalence is established.

8. Suppose the person A_j knows j people where $j = 1, 2, \dots, n-1$. We claim that the remaining person A_n knows $\left\lfloor \frac{n}{2} \right\rfloor$ people.

When $n = 2$, A_1, A_2 should know each other. So A_2 knows 1 person. When $n = 3$, A_2 knows both A_1 and A_3 , while A_1 only knows A_2 . Hence, A_3 knows exactly 1 person.

Assume the claim is true for all $n = 2, 3, \dots, k$ where $k \geq 3$. When $n = k+1$, since A_k knows k people, he must know all the others. Then the only person known by A_1 is A_k . By removing A_1 and A_k , there are $k-1 \geq 2$ people left, where $A_2, A_3, \dots, A_{k-1}$ know $1, 2, \dots, k-2$ people respectively (not including A_k). By the inductive hypothesis, A_{k+1} knows exactly $\left\lfloor \frac{k-1}{2} \right\rfloor$ people among $A_2, A_3, \dots, A_{k-1}$. Together with A_k , he knows exactly $\left\lfloor \frac{k+1}{2} \right\rfloor$ people. By induction, the claim is true for any $n \geq 2$.

Therefore, the answer is $\left\lfloor \frac{n}{2} \right\rfloor$.

9. Suppose on the contrary that $P_1 - P_2 - \dots - P_k - P_1$ is a closed polygonal line. WLOG we may assume P_1P_2 has the smallest distance among all distances formed by these k points. As P_2 and P_3 are connected, and P_2 is closer to P_1 than P_3 , the point P_2 must be the nearest point to P_3 . Inductively, we can show that P_m is the nearest point to P_{m+1} . Thus, P_{k-1} is the nearest point to P_k . However, P_2 is the nearest point to P_1 . This means P_k and P_1 cannot be connected. This is a contradiction. Hence, there is no closed polygonal line.

10. We use terminologies in graph theory. It is trivial if it is a complete graph. Assume there are 2 vertices A and B which are not adjacent. Let $\mathcal{A}$ and $\mathcal{B}$ be the set of vertices which are adjacent to A and B respectively. Suppose on the contrary that $|\mathcal{A}| \neq |\mathcal{B}|$. WLOG assume $|\mathcal{A}| > |\mathcal{B}|$.

For each $V \in \mathcal{A}$, there exists a unique W which is adjacent to both V and B . Then $W \in \mathcal{B}$. We define a mapping $f : \mathcal{A} \rightarrow \mathcal{B}$ which sends V to W . This is well-defined as W is unique. As $|\mathcal{A}| > |\mathcal{B}|$, there must be two distinct vertices $V_1, V_2 \in \mathcal{A}$ such that $f(V_1) = f(V_2) = W$ for some $W \in \mathcal{B}$. Note that $W \neq A$ since W is adjacent to B . Thus, both W and A are adjacent to the vertices V_1 and V_2 . This contradicts the given condition. Therefore, we must have $|\mathcal{A}| = |\mathcal{B}|$.

11. We use terminologies in graph theory. Let the vertices be $V_1, V_2, \dots, V_{2n}$. Since the graph is connected, we can find a path from V_1 to V_2 , a path from V_3 to V_4 , $\dots$, and a path from V_{2n-1} to V_{2n} . Consider a set S of edges as follows. An edge belongs to S if and only if it belongs to an odd number of the above paths. We prove that each vertex is the endpoint of an odd number of edges in S .

Consider V_1 . It is the endpoint of exactly one path. Each time when it appears as an intermediate vertex in a path, there must be two edges incident to V_1 . If the same edge appears twice in the paths, they cancel out each other and hence the parity of the number of edges incident to V_1 is unchanged. So there must be an odd number of edges in S incident to V_1 . By symmetry, the same holds for any other vertex. So we can remove all edges except the edges in S .

12. Using terminologies in graph theory, it means a simple graph is bipartite if and only if it does not contain any odd cycle.

If the graph is bipartite, we partition the graph into the groups $\mathcal{A}$ and $\mathcal{B}$. Consider any cycle $V_1 - V_2 - \dots - V_k - V_1$. WLOG assume $V_1 \in \mathcal{A}$. Then $V_2 \in \mathcal{B}$, $V_3 \in \mathcal{A}$ and so on. Since V_k and V_1 are adjacent, we must have $V_k \in \mathcal{B}$, and hence k is even from the above pattern. So any cycle must have even length.

If the graph does not contain any odd cycle, we partition the vertices as follows. WLOG assume the graph is connected. We put an arbitrary vertex V in $\mathcal{A}$. Then for any other vertex W , there exists a path from V to W from the connectedness. We put W in $\mathcal{A}$ if the length of this path is even. Otherwise, we put W in another group $\mathcal{B}$. Note that this uniquely determines the group to which W belongs. Indeed, if there are two paths $\mathbf{a}$ and $\mathbf{b}$ from V to W , then $V \xrightarrow{\mathbf{a}} W \xrightarrow{\mathbf{b}^{-1}} V$ is a cycle (where $\mathbf{b}^{-1}$ is the path from W to V obtained by reversing the steps in $\mathbf{b}$). This cycle must have even length, and hence the lengths of $\mathbf{a}$ and $\mathbf{b}$ have the same parity.

It is not hard to see that $\mathcal{A}$ and $\mathcal{B}$ satisfy the bipartite condition. Suppose on the contrary that two vertices in the same group are adjacent. Let V_1 and V_2 be two such vertices.

We find a path $\mathbf{a}_1$ from V to V_1 , and a path $\mathbf{a}_2$ from V to V_2 . Then $V \xrightarrow{\mathbf{a}_1} V_1 - V_2 \xrightarrow{\mathbf{a}_2^{-1}} V$

is a cycle of odd length since the lengths of $\mathbf{a}_1$ and $\mathbf{a}_2$ have the same parity. This is a contradiction. So the equivalence is established.

13. The answer is 1979.

We use terminologies in graph theory. Firstly, suppose there exists a vertex A with degree ≤ 1979 . This means A is not adjacent to two vertices B and C . For any other vertex D , by considering the four vertices A, B, C, D , the edges AD, BD, CD must exist in order that the condition is satisfied. Also, for any other vertices D, E , by considering the four vertices A, B, D, E , the edge DE must exist (otherwise, none of A, B, D, E can be adjacent to all the other three vertices). These show every vertex other than A, B, C is adjacent to all other vertices. Hence, the number of vertices with degree 1981 is 1979 in this case. The condition is satisfied.

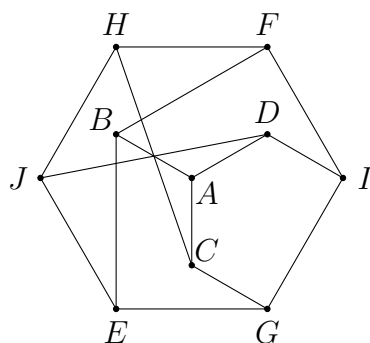
Secondly, suppose all vertices have degree at least 1980. We may assume there exists a vertex A with degree 1980. Suppose A is not adjacent to B . Then the degree of B is also 1980. Hence, all other vertices are adjacent to both A and B . Also, for any other vertices C and D , by considering the four vertices A, B, C, D , the edge CD must exist. This shows every vertex other than A, B is adjacent to all other vertices. Hence, the number of vertices with degree 1981 is 1980 in this case.

The two cases show that the minimum number is 1979.

14. The answer is 10.

We use terminologies in graph theory. A vertex A is connected to at most 3 other vertices B, C, D . Each of B, C, D (if exists) is connected to at most 2 other vertices apart from A . There cannot be more vertices since otherwise the distance between A and such a vertex is at least 3. This shows the number of vertices is at most $1 + 3 + 2 \times 3 = 10$.

We construct a case with exactly 10 vertices as shown below. Each vertex has degree 3 and the distance between any 2 vertices is at most 2 (indeed, it suffices to check the vertices A, B, E due to symmetry).



15. The answer is $n - 2$. We use terminologies in graph theory. Suppose we remove an edge AB from a complete graph of n vertices. Then it is obvious that all the $n - 2$ vertices except A and B are popular.

Since the vertex with the minimum degree cannot be popular, there cannot be n popular vertices. Now, suppose there are exactly $n - 1$ popular vertices. Then there is a unique vertex A with minimum degree d_1 . Let d_2 be the second smallest degree. Suppose B is one of the k vertices which has degree d_2 . Note that B must be adjacent to A , since otherwise the assumption cannot hold. As this holds for any vertex B with degree d_2 and the degree of A is d_1 , we have $k \leq d_1$.

Suppose B is adjacent to x vertices with degree d_2 . Then the average degrees of the neighbours of B is

$$\geq \frac{d_1 + xd_2 + (d_2 - 1 - x)(d_2 + 1)}{d_2} = d_2 + \frac{d_1 - 1 - x}{d_2} \geq d_2 + \frac{k - 1 - x}{d_2} \geq d_2.$$

This is a contradiction. So there are at most $n - 2$ popular vertices.

16. Using terminologies in graph theory, we have a bipartite graph with the partition $\mathcal{A}$ and $\mathcal{B}$ such that each vertex in $\mathcal{A}$ is adjacent to at least 4 vertices in $\mathcal{B}$ and every pair of vertices in $\mathcal{A}$ shares a common neighbour in $\mathcal{B}$. We need to colour the vertices in $\mathcal{B}$ black and white such that each vertex in $\mathcal{A}$ is adjacent to a black vertex and a white vertex.

If there exists $V \in \mathcal{B}$ which is adjacent to all vertices in $\mathcal{A}$, then we can colour V in black and all other vertices in white. The condition clearly holds.

If not, by choosing any $W_1, W_2 \in \mathcal{A}$ which are adjacent to $V_1 \in \mathcal{B}$, there must be $W_3 \in \mathcal{A}$ which is not adjacent to V_1 . From the condition, W_1, W_3 must be adjacent to some vertex $V_2 \in \mathcal{B}$. Similarly, W_2, W_3 must be adjacent to some vertex $V_3 \in \mathcal{B}$. Note that V_1, V_2, V_3 are pairwise distinct from the assumption and the condition. We colour V_1, V_2, V_3 and any vertex in $\mathcal{B}$ which is not adjacent to all of W_1, W_2, W_3 in black, and we colour the remaining vertices in white. We claim that this colouring satisfies the required condition.

Firstly, W_1 is adjacent to the black vertex V_1 and at least one vertex distinct from V_1, V_2, V_3 . Such a vertex must be white from the construction. Similarly, we can check for W_2 and W_3 . Secondly, for any other $W_4 \in \mathcal{A}$, it cannot be adjacent to two of V_1, V_2, V_3 . Otherwise, W_4 has more than one common neighbour with one of W_1, W_2, W_3 . WLOG assume W_4 is not adjacent to V_2 and V_3 . Then W_3 and W_4 are adjacent to a white vertex. Also, W_4 is adjacent to a vertex which is not a common neighbour with W_1, W_2 or W_3 . Such a vertex must be black from the construction. These show the construction works.

17. (a) We use terminologies in graph theory. If A designates B as her best friend, then we draw a directed edge from A to B . If A is an n -best friend, then we say that A is n -good. We first claim that if a vertex is n -good, then it is k -good for any positive integer $k \leq n$. We prove this by induction on n . The base case $n = 1$ is trivial. Consider the inductive step.

Suppose A is an n -good vertex. By definition, there exists an edge $B \rightarrow A$ where B is an $(n-1)$ -good vertex. By the inductive hypothesis, B is k -good for any $k = 1, 2, \dots, n-1$. Therefore, A is k -good for $k = 2, 3, \dots, n$. Also, as we have the edge $B \rightarrow A$, we know that A is also 1-good. So the inductive step is established. The claim follows by induction.

Now, suppose on the contrary that A is popular but there isn't a popular vertex B such that $B \rightarrow A$. Since there is a finite number of edges directed to A , we may let $B_1, B_2, \dots, B_m$ be the only vertices such that the edge $B_j \rightarrow A$ exists for each j . From the assumption and the above claim, for each vertex B_j , there exists a largest positive integer f_j such that B_j is k -good for $k = 1, 2, \dots, f_j$ but not for other k 's. Let $M = \max\{f_1, f_2, \dots, f_m\}$. Then A is not $(M+2)$ -good by definition, since none of the B_j 's is $(M+1)$ -good. This is a contradiction. So every popular person is the best friend of a popular person.

- (b) It suffices to provide a counterexample. Let X, Y, Z and V_{ij} be some vertices where $i \geq j \geq 1$. Suppose we have the edges

$$V_{i1} \rightarrow V_{i2} \rightarrow \dots \rightarrow V_{ii} \rightarrow X$$

for each $i = 1, 2, \dots$. Also, suppose we have the edges $X \rightarrow Y, Y \rightarrow Z$ and $Z \rightarrow Y$. It is clear that each V_{ij} with $j \geq 2$ is a $(j-1)$ -good vertex, and hence X is i -good for any positive integer i . However, none of the V_{ii} 's is popular.

18. We use terminologies in graph theory. We count the number of pairs $(A, \{B, C\})$ such that A, B, C are distinct vertices and both AB and AC are edges.

Suppose there are k vertices with degree 10. Then the other $n-k$ vertices have degree

11. For each A with degree d , there are $\binom{d}{2}$ ways to choose the set $\{B, C\}$ such that both AB and AC are edges. So the number of pairs is

$$k \binom{10}{2} + (n-k) \binom{11}{2} = 45k + 55(n-k) = 55n - 10k.$$

For each of the $\binom{n}{2}$ sets $\{B, C\}$, there are exactly 5 choices of A . So the number of pairs is $5 \binom{n}{2}$. This implies

$$55n - 10k = \frac{5n(n-1)}{2}.$$

Using $k \geq 0$, we get $55n \geq \frac{5n(n-1)}{2}$, i.e. $n \leq 23$.

Using $k \leq n$, we get $45n \leq \frac{5n(n-1)}{2}$, i.e. $n \geq 19$.

19. Using terminologies in graph theory, we have a complete graph with directed edges. We need to find an upper bound for the number of directed triangles. Note that for any given triangle, it either forms a cycle or there exist two edges having the same initial vertex as well as two edges having the same terminal vertex. We count the number of triangles of the latter kind. Consider those triples (A, B, C) of vertices such that the edges are $A \rightarrow B$, $A \rightarrow C$ and $B \rightarrow C$.

Let $d_1, d_2, \dots, d_n$ be the in-degrees of the vertices. For a vertex with in-degree d , there are $\binom{d}{2}$ triples for which this vertex is C . Also, since its out-degree is $n - 1 - d$, there are $\binom{n - 1 - d}{2}$ triples for which this vertex is A . Note that each triple is counted twice if we sum over the contributions from all vertices. Hence, there are

$$\frac{1}{2} \sum_{k=1}^n \left(\binom{d_k}{2} + \binom{n - 1 - d_k}{2} \right)$$

triangles which are not cycles. By Jensen's inequality, we have

$$\binom{d}{2} + \binom{n - 1 - d}{2} \geq 2 \binom{\frac{d + (n - 1 - d)}{2}}{2} = \frac{(n - 1)(n - 3)}{4}.$$

It follows that the number of directed triangles is

$$\begin{aligned} \binom{n}{3} - \frac{1}{2} \sum_{k=1}^n \left(\binom{d_k}{2} + \binom{n - 1 - d_k}{2} \right) &\leq \frac{n(n - 1)(n - 2)}{6} - \frac{n(n - 1)(n - 3)}{8} \\ &= \frac{1}{4} \binom{n + 1}{3}. \end{aligned}$$

20. The answer is 3780.

We use terminologies in graph theory. The edges are coloured in 5 colours. We need to find the smallest possible number of pairs $(A, \{B, C\})$ such that the edges AB and AC have the same colour. Consider each of the vertices V . Let there be $a_1, a_2, \dots, a_5$ edges of the 5 colours incident to V respectively. Note that $a_1 + a_2 + \dots + a_5 = 35$. The number of pairs with $A = V$ is

$$\binom{a_1}{2} + \binom{a_2}{2} + \dots + \binom{a_5}{2} \geq 5 \binom{\frac{a_1 + a_2 + \dots + a_5}{5}}{2} = 105$$

by Jensen's inequality. It follows that there are at least $36 \times 105 = 3780$ pairs as each pair is counted once in this way.

We now provide a graph for which there are exactly 3780 pairs. Let the vertices be $V_1, V_2, \dots, V_{36}$. We put the vertices in a way that $V_1 V_2 \dots V_{35}$ is a regular polygon and V_{36} is its centre. We partition the edges as follows. Each group consists of the edges

which are parallel to a side of the polygon together with the edge joining V_{36} and the vertex which has not paired up. This gives 35 groups of edges such that every edge belongs to exactly one group and the 18 edges in each group covers all 36 vertices exactly once.

We use each of the 5 colours to colour the edges in exactly 7 groups. Then $a_j = 7$ for each j , and hence equality in the above estimation holds. So there are exactly 3780 pairs.

Ramsey-typed Problems

21. Among the 5 segments joining P_1 , there are $\left\lceil \frac{5}{2} \right\rceil = 3$ of them having the same colour by pigeonhole principle. WLOG assume the segments joining P_1 and one of P_2, P_3, P_4 are black. If any of the segments P_2P_3 , P_3P_4 and P_4P_2 is black, say P_2P_3 , then $P_2P_3P_4$ is a black triangle. In not, then all of these segments are white, in which case we have a white triangle $P_2P_3P_4$.
22. We use vertices to denote the irrational numbers. Two numbers a and b are joined by an edge if and only if $a + b$ is irrational.

Since the Ramsey number $R(3, 3)$ is equal to 6, either there exists a triangle or there exist three numbers such that no two of them are joined by an edge. The former case is exactly what we want. Suppose the latter case holds. Then we can find a, b, c such that $a + b$, $b + c$ and $c + a$ are rational. It follows that

$$a = \frac{(a + b) + (c + a) - (b + c)}{2}$$

is rational, which is a contradiction. Therefore, this case cannot hold and we are done.

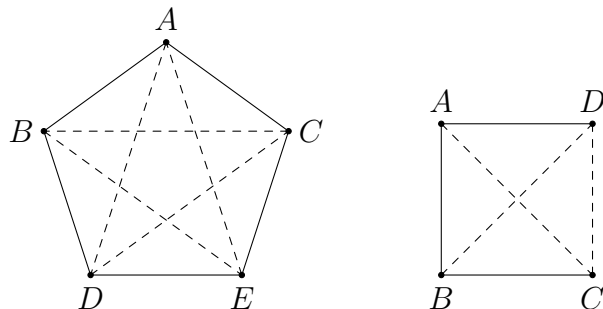
23. The answer is 4.

We use terminologies in graph theory. The cities are represented by vertices, and we colour the edges of the complete graph red and blue depending on the companies operating the flight. The condition means that there is no monochromatic simple cycle. Since the Ramsey number $R(3, 3)$ is equal to 6, there must be a monochromatic triangle if $n \geq 6$. So we have $n \leq 5$.

For $n = 5$, for each vertex A , if there are 3 edges with A as an endpoint which are of the same colour, then we can apply the proof of $R(3, 3) = 6$ to get a monochromatic triangle. So we may assume among the 4 edges with A as an endpoint, 2 are red and 2 are blue. This holds for any other vertex.

WLOG assume AB and AC are red. Note that BC cannot be red, or otherwise the cycle $A - B - C - A$ is red. So we may assume BD is red for some other vertex D . Similarly, if CD is red, then $A - B - D - C - A$ is a red cycle. So CD is blue, and the

remaining vertex E must be such that CE is red. In order that there are 2 red edges directed from each vertex, DE must be red. Then $A - B - D - E - C - A$ is a red cycle. Therefore, we cannot have $n = 5$.



It is possible to have $n = 4$. For example, if DA, AB, BC are red while AC, BD, CD are blue, then it is routine to check that there is no monochromatic cycle. So the maximum n is 4.

24. Using terminologies in graph theory, we have a complete graph of 17 vertices, where every edge is painted by one of the three colours red, blue or green. We need to show that there exists a monochromatic triangle.

Pick an arbitrary vertex A . Among the 16 edges incident to A , there are $\left\lceil \frac{16}{3} \right\rceil = 6$ of them having the same colour by the pigeonhole principle. WLOG assume the edges joining A and one of B, C, D, E, F, G are red. If any edge joining two of B, C, D, E, F, G is red, say $B - C$, then we are done since ABC is a red triangle. Thus, we may assume any such edge is blue or green. Since the Ramsey number $R(3, 3)$ is equal to 6, there must be a monochromatic triangle among these 6 vertices. Hence, we are done in every case.

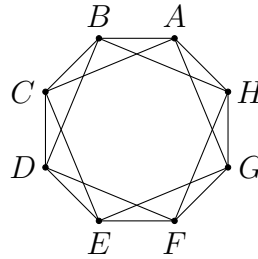
25. We use terminologies in graph theory. We join 2 vertices if the 2 mathematicians speak a common language. We colour the edges such that different colours represent different languages.

If the graph is a complete graph, then any vertex A has degree 8. As the edges incident to A are in 3 colours only, there are 2 edges from A , say AB and AC , having the same colour. Then A, B, C speak the same language.

If the graph is not a complete graph, we may assume AB is not an edge. Consider the group A, B, C for any other vertex C . Then at least one of AC and BC is an edge. As there are 7 choices for C , by the pigeonhole principle, we may assume $\left\lceil \frac{7}{2} \right\rceil = 4$ other vertices are adjacent to A . The 4 corresponding edges incident to A are in 3 colours only. So there are 2 of them which have the same colour. Again, this corresponds to 3 mathematicians speaking the same language.

26. The answer is 8. This is to determine the Ramsey number $R(3, 4)$ and then n should be $R(3, 4) - 1$.

We use terminologies in graph theory. Consider the following graph with 8 vertices. We construct a simple cycle $A - B - C - D - E - F - G - H - A$ and the squares $ACEG$ and $BDFH$. For any 3 vertices, there must be 2 of them which are neighbours in the cycle or there is exactly one other vertex in between them in the cycle. So they are adjacent to each other. For any 4 vertices, we may assume A is one of them. Note that A is only adjacent to B, C, G, H . As B and G are not adjacent, C and H are not adjacent, we cannot find 3 more vertices from B, C, G, H such that any 2 of them are adjacent. So the conditions are satisfied.



Suppose there are at least 9 vertices. It suffices to consider 9 vertices. For a vertex A , if $\deg(A) \geq 6$, then since $R(3, 3) = 6$, there must be 3 vertices among those adjacent to A such that they are pairwise adjacent or pairwise non-adjacent. In the former case, these 3 vertices together with A form a K_4 , which is a contradiction. In the latter case, the condition is violated. So we may assume $\deg(A) \leq 5$ for any vertex A .

If $\deg(A) \leq 4$, we suppose A is not adjacent to B, C, D, E . If B and C are not adjacent, then the group A, B, C violates the condition. Otherwise, we may assume any 2 of B, C, D, E are adjacent due to symmetry. Then this gives a K_4 , contradicting the given condition. Hence, we may assume $\deg(A) = 5$ for any vertex A .

Recall that the number of edges is half the sum of degrees of all vertices. So the number of edges is $\frac{5 \times 9}{2}$, which is not an integer. This is impossible. Thus, the maximum value of n is 8.

27. (1) We use terminologies in graph theory. If there are 3 vertices A, B, C which are pairwise not adjacent, then by considering any group A, B, C, D, E , there must be a triangle. WLOG we may assume the triangle is ADE . This means D and E are adjacent for any other vertices D and E . In that case, we can simply pick 4 vertices other than A, B, C .

If for any 3 vertices, there are 2 of them which are adjacent, we consider the following cases.

- The degree of A is at most 3 for some vertex A .

Suppose A is not adjacent to B, C, D, E . Then by considering A, B, C , the vertices B and C must be adjacent from our assumption. Due to symmetry, any two vertices among B, C, D, E are adjacent. So we are done.

- The degree of A is at least 5 for some vertex A .

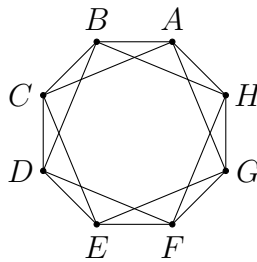
Suppose A is adjacent to B, C, D, E, F . Then by considering B, C, D, E, F , we can find a triangle, say BCD . Then A, B, C, D are pairwise adjacent.

- The degree of every vertex is exactly 4.

Suppose A is adjacent to B, C, D, E and not adjacent to F, G, H . Then by considering A, F, G , the vertices F and G must be adjacent. Similarly, FGH is a triangle. If any two vertices among B, C, D, E are adjacent, then we are done. So we may assume B and C are not adjacent. If one of them is adjacent to all of F, G, H , then we are done by choosing this vertex together with F, G, H . So we may assume each of B, C is not adjacent to one of F, G, H .

By considering B, C, F , we know that F must be adjacent to B or C . Similarly, it is not possible for B and C to be not adjacent to the same vertex among F, G, H . WLOG assume B and F are not adjacent, C and G are not adjacent. If we put the vertices B, F, A, G, C in a circle in that order, then any two consecutive vertices are not adjacent. However, there exists a triangle among these vertices from the condition. Such a triangle must contain two consecutive vertices in the circle. This is a contradiction. So we are done in this case, and the proof is complete.

- (2) Not always true. A counterexample is given as follows. Firstly, we construct a simple cycle $A - B - C - D - E - F - G - H - A$ and the squares $ACEG$ and $BDFH$. Consider any 6 vertices. WLOG assume A is not one of them. Then there are 5 of them chosen from B, C, D and E, F, G . By the pigeonhole principle, one of the triangles BCD and EFG must exist. So the condition is satisfied.

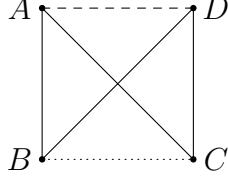


However, for any 4 vertices, there must be 2 of them whose distance in the cycle is at least 3. So they are not adjacent to each other.

28. The answer is 4.

We use terminologies in graph theory. We have a complete graph of n vertices such that each edge is coloured with red, blue or green. All colours have been used. For each vertex, the edges incident to it are in at most 2 different colours. Also, there is no triangle whose 3 edges are of the same colour.

Firstly, consider a graph with 4 vertices A, B, C, D such that the edges AB, AC, BD, CD are red, AD is blue and BC is green. It is routine to check that all conditions are satisfied.



We now prove that $n \leq 4$. Suppose on the contrary that $n \geq 5$. We say that a vertex is of type RB if there is no green edge connected to it. Similarly, we define vertices of type RG and BG. If a vertex may belong to more than one type, then we arbitrarily assign such a type for it. Firstly, we note there are at most 2 types of vertices. Indeed, suppose there exist vertices A, B, C such that A is of type RB, B is of type RG and C is of type BG. We may further find a vertex D which is of type RB, say. Then both AB and BD are red, both AC and CD are blue. As AD is either red or blue, we must have a red triangle ABD or a blue triangle ACD . This contradicts the given condition.

WLOG assume the vertices are of type RB and type RG only. By the pigeonhole principle, since $n \geq 5$, there are 3 vertices A, B, C of type RB, say. Let D be a vertex of type RG. Then each of the edges AD, BD, CD must be red. If one of AB, BC, CA is red, then we get a red triangle. If not, then all of them are blue, and we get a blue triangle. So there are contradictions in both cases.

Hence, $n \leq 4$ and the maximum number of vertices is 4.

29. We prove this by induction. If $m = 2$, then for any simple graph with $\binom{n}{1} = n$ vertices, there are either 2 adjacent vertices or all the n vertices are pairwise non-adjacent. So $R(2, n) \leq n$. Similarly, the case $n = 2$ holds.

Consider the inductive step (m, n) with $m, n \geq 3$. Assume the bound holds for any case (m', n') with $m' \leq m$, $n' \leq n$ and $(m', n') \neq (m, n)$. Consider any vertex A in a simple graph with $k = \binom{m+n-2}{m-1}$ vertices. Among the other $k-1$ vertices, if $\binom{m+n-3}{m-2}$ of them are adjacent to A , then by the inductive hypothesis, either there is a K_{m-1} among these vertices or there are n vertices which are pairwise non-adjacent. We are done in the latter case. In the former case, those $m-1$ vertices together with A form a K_m . So the assertion holds in this case.

If $\binom{m+n-3}{m-1}$ of the other $k-1$ vertices are not adjacent to A , then by the inductive hypothesis, either there is a K_m among these vertices or there are $n-1$ vertices which are pairwise non-adjacent. We are done in the former case. In the latter case, those $n-1$

vertices together with A form a group of n vertices which are pairwise non-adjacent. So the assertion holds in this case.

If none of the above holds, then we have

$$\left(\binom{m+n-3}{m-2} - 1 \right) + \left(\binom{m+n-3}{m-1} - 1 \right) \geq k - 1 = \binom{m+n-2}{m-1} - 1.$$

However, we also have the identity

$$\binom{m+n-3}{m-2} + \binom{m+n-3}{m-1} = \binom{m+n-2}{m-1}.$$

This is a contradiction. Hence, $R(m, n) \leq \binom{m+n-2}{m-1}$ by induction.

Extremal Case

30. Consider all partitions of the n people into 2 groups. For each partition, we assign a number to each person which is the number of people he knows in the same group. Then we choose a partition which minimizes the sum of the numbers assigned. We claim that the goal is met in this partition.

Suppose on the contrary that a person A knows at least 2 people in the same group X . Then A knows at most 1 person in the other group Y . We consider the partition by putting A to group Y . Then the number assigned to A is decreased by at least 1, the number assigned to each person in group X who knows A is decreased by 1, and the number assigned to the person (who may not exist) in group Y who knows A is increased by 1. Thus, the sum is decreased by at least 2. This contradicts the minimality of the choice.

Hence, every person knows at most 1 person in the same group.

31. **Solution 1:**

We use terminologies in graph theory. We consider a matching with the maximum number of pairs. Suppose on the contrary that there are $k \leq 5$ pairs. Then the remaining $20 - 2k$ vertices are pairwise non-adjacent. From the condition, there are at least $20 - 2k$ more edges incident to one of these vertices. Together with the k edges in the matching, there are at least $(20 - 2k) + k = 20 - k \geq 15$ edges. This contradicts the given condition that there are exactly 14 edges.

Solution 2:

There are 28 members involved in the games counted with multiplicities. As there are 20 members, there are at most $28 - 20 = 8$ games for which one member has played before when he plays this game. Then there are at least $14 - 8 = 6$ games for which both members have not played before. Then any such 6 games must consist of 12 distinct players.

32. We use terminologies in graph theory. Suppose there exists an Eulerian cycle (a cycle passing through every edge exactly once). Consider any vertex V . If there are k copies of V in this cycle, then there are $2k$ edges incident to V since there are 2 edges incident to each such copy. This shows the degree of each vertex is even.

Suppose the degree of each vertex is even. Consider a longest path $\mathbf{c}$ without repeated edges, say from V_1 to V_2 . Suppose we have travelled to V_2 for m times. If $V_2 \neq V_1$, then we have travelled exactly $2m - 1$ edges incident to V_2 as above. However, since the degree of V_2 is even, there exists another edge incident to V_2 for which we have not yet travelled. This is a contradiction since otherwise the path is not the longest. This shows $V_2 = V_1$ and $\mathbf{c}$ is indeed a cycle.

Next, assume $\mathbf{c}$ does not consist of all edges. By removing all edges in $\mathbf{c}$, we are left with a graph such that the degree of each vertex is even. Using the same argument, we can find a cycle $\mathbf{c}'$ in this subgraph passing through a vertex V which also belongs to $\mathbf{c}$. If we travel from V along $\mathbf{c}$ and return to V , and then travel along $\mathbf{c}'$ and return to V again, we form a longer cycle without repeated edges. This is a contradiction. So $\mathbf{c}$ is an Eulerian cycle.

33. We use terminologies in graph theory. Let $V_1 - V_2 - \dots - V_k$ be the longest simple path in the graph. Since $\deg(V_k) \geq 3$, the vertex V_k is adjacent to 2 more vertices other than V_{k-1} . From the maximality of the length of the path, these 2 vertices must be some of the V_j 's. Suppose V_k is adjacent to V_i and V_j where $i < j$. Consider the paths

$$\begin{aligned} V_i - V_{i+1} - \dots - V_{k-1} - V_k, \\ V_i - V_{i+1} - \dots - V_j - V_k. \end{aligned}$$

If one of these has an odd length, then it can be extended to a cycle of even length by connecting the edge $V_i V_k$. If both have even lengths, then $V_j - V_{j+1} - \dots - V_k - V_j$ is a cycle of even length. This completes the proof.

34. We use terminologies in graph theory. Suppose on the contrary that there is no Hamiltonian cycle (a simple cycle of length $2n$). Consider such a graph with the greatest number of edges. Clearly, the graph is not the complete graph. Assume $V_1 V_{2n}$ is not an edge. If we add this edge, then there should be a Hamiltonian cycle from the maximality of the choice. So we may assume the vertices form a path $V_1 - V_2 - \dots - V_{2n}$.

As $\deg(V_1) \geq n$, the vertex V_1 is adjacent to at least $n-1$ of the vertices $V_3, V_4, \dots, V_{2n-1}$. Similarly, V_{2n} is adjacent to at least $n-1$ of the vertices $V_2, V_3, \dots, V_{2n-2}$. As there are only $2n-3$ pairs (V_{j+1}, V_j) where $j = 2, 3, \dots, 2n-2$, and $(n-1) + (n-1) > 2n-3$, there must be a pair (V_{j+1}, V_j) such that V_1 is adjacent to V_{j+1} , and V_{2n} is adjacent to V_j . Thus, we can find a cycle $V_1 - V_2 - \dots - V_j - V_{2n} - V_{2n-1} - \dots - V_{j+1} - V_1$ of length $2n$. This is a contradiction. So any such graph must contain a Hamiltonian cycle.

35. We use terminologies in graph theory. Let the largest clique in the graph have k vertices. We put these k vertices in one group $\mathcal{S}$. It suffices to show that the remaining vertices are isolated.

Suppose on the contrary that two vertices A and B not in $\mathcal{S}$ are adjacent. If A is adjacent to every vertex in $\mathcal{S}$, then $\{A\} \cup \mathcal{S}$ is a clique of size $k + 1$, which is a contradiction. So A is not adjacent to some vertex C in $\mathcal{S}$. Similarly, we can find $D \in \mathcal{S}$ which is not adjacent to B .

If $C \neq D$, then among the vertices A, B, C, D , any 3 of them must include the pair (A, B) or the pair (C, D) . Since A, B are adjacent and C, D are adjacent, those 3 vertices must be pairwise adjacent. However, there is a pair among $(A, C), (A, D), (B, C), (B, D)$. This is a contradiction since none of these forms an edge.

Therefore, we must have $C = D$, and this holds for any choice of C and D . This means all vertices in $\mathcal{S}$ other than C are adjacent to both A and B . In that case, we can replace C by A and B to form a clique of size $k + 1$. This is again a contradiction.

So the vertices not in $\mathcal{S}$ are isolated and we are done.

36. We use terminologies in graph theory. Let the largest clique in the graph have k vertices. We prove that $k \geq n + 1$. Suppose on the contrary that $k \leq n$. Consider any group of n vertices including these k vertices. From the given condition, there exists another vertex which is adjacent to all these n vertices. In particular, it is adjacent to all the k vertices in the clique. This gives a clique of size $k + 1$, which is a contradiction.

Now, consider any clique of size $n + 1$. For the other n vertices not in this clique, there exists a vertex in this clique which is adjacent to all those n vertices from the given condition. So this vertex is adjacent to all other vertices, and we are done.

37. Using terminologies in graph theory, we have a complete graph with directed edges such that the out-degree of each vertex is at least 1. We need to prove the existence of a directed cycle of length 3.

Let A be a vertex with the minimum out-degree. From the condition there exists a vertex B such that $A \rightarrow B$. For any vertex C such that $C \rightarrow A$, if $B \rightarrow C$ then we are done since $A \rightarrow B \rightarrow C \rightarrow A$ is a cycle. So we may assume any C for which $C \rightarrow A$ satisfies $C \not\rightarrow B$. Together with the edge $A \rightarrow B$, we find that the in-degree of A is smaller than that of B . Thus, the out-degree of A is larger than that of B , which contradicts the choice of A . Therefore, we can always find C such that $C \rightarrow A$ and $B \rightarrow C$.

38. Using terminologies in graph theory, we have a simple graph of n vertices which does not contain any triangle and is not bipartite. We need to prove the existence of a vertex with degree at most $\frac{2}{5}n$. Note that the given condition also implies $n \geq 3$.

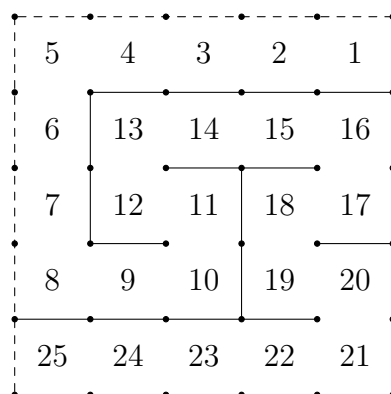
There must be an odd cycle since the graph is not bipartite. We choose such a cycle $V_1 - V_2 - \cdots - V_k - V_1$ with the smallest length. From the condition, $k \geq 5$. Note that no two of these vertices are adjacent unless they are neighbours in the cycle. Indeed, suppose V_i and V_j are adjacent. Then one of the cycles $V_i - V_{i+1} - \cdots - V_j - V_i$ and

$V_j - V_{j+1} - \dots - V_i - V_j$ has odd length since their sum of lengths is $k + 2$, which is odd. However, this odd cycle has length $< k$, which contradicts the minimality of k .

For any other vertex W , we claim that W is adjacent to at most 2 vertices among the V_j 's. From the given condition, W cannot be adjacent to two neighbouring vertices in the cycle. Therefore, if W is adjacent to at least 3 of the vertices, then it is adjacent to 2 of them which are at least 3 steps apart in the cycle. Note that this also requires the fact that k is odd (indeed, the case $k = 6$ and W is adjacent to V_1, V_3, V_5 should be rejected). Let V_i and V_j be two such vertices. Then one of the cycles $V_i - V_{i+1} - \dots - V_j - W - V_i$ and $V_j - V_{j+1} - \dots - V_i - W - V_j$ has odd length and its length must be less than k . This is a contradiction. So any other vertex is adjacent to at most 2 of the V_j 's.

From above, the total degrees of the V_j 's is at most $2k + 2(n - k) = 2n$. By the pigeonhole principle, there exists a vertex V_j with degree $\leq \frac{2n}{k} \leq \frac{2n}{5}$.

39. Consider a simple graph as follows. The vertices are the endpoints of the edges of the unit squares. Two vertices are joined by an edge if and only if the segment joining these endpoints separates two numbers with difference greater than 1. Consider a longest path of the simple graph. There are a few possibilities.



- If one of the endpoints of the path meets the path again at some vertex, then there is a cycle within this path. This cycle separates the board into two disconnected regions. However, we can always find a number in one region which is consecutive to a number in another region since the numbers in the whole board are consecutive. Then these numbers are adjacent and they should lie in two squares which are not separated by an edge. This is a contradiction.
- If both endpoints of the path meet the boundary of the board, then the path separates the board into two disconnected regions. Again, this yields a contradiction.
- For the remaining cases, one of the endpoints of the path must neither meet the path again nor the boundary of the board. Due to the maximality of the length

of the path, such an endpoint P must have degree 1 and it is surrounded by 4 squares. Let the numbers in those 4 squares be a, b, c, d in the clockwise direction such that a and d are not consecutive. Then we must have $b = a \pm 1$, $c = a \pm 2$ and $d = a \pm 3$. This is what we want.

40. In other words, the number of edges in a maximum matching is equal to the number of vertices in a minimum vertex cover.

Since at least one vertex in each of the M edges in the matching should be chosen to form a vertex cover, we must have $N \geq M$. Suppose $N > M$ for some bipartite graph. We choose such a graph G with the minimum number of vertices. If there is more than one such graph, we choose one of them which has the smallest number of edges.

Clearly, we may assume G is connected. We claim that there exists a vertex with degree at least 3. Suppose not. Starting from a vertex V_1 , suppose $V_1 - V_2$ is an edge (as G is connected). Since $\deg(V_2) \leq 2$, we can find at most one other vertex V_3 which is adjacent to V_2 . Continuing in this way, we can form a longer path $V_1 - V_2 - \dots - V_k$ until we reach a vertex V_k with degree 1 or V_k is adjacent to V_1 . Note that there are no other vertices as G is connected. In the previous case, G is a single path with k vertices.

Then it is not hard to see that $M = \left\lfloor \frac{k}{2} \right\rfloor = N$. In the latter case, G is a single cycle

with k vertices, where k is even since G is bipartite. Then $M = \frac{k}{2} = N$.

Let A be a vertex such that $\deg(A) \geq 3$. Suppose B, C, D are adjacent to A . If every matching with M edges consists of the vertex B , then by removing B , the number of edges in a maximum matching is $M - 1$. From the choice of G , we can choose $M - 1$ vertices other than B in G which cover all edges, possibly except those incident to B . Together with B , we have a vertex cover with M vertices. This contradicts $N > M$.

Therefore, there exists a matching with M edges that does not consist of B . WLOG assume AC is not in the matching. By removing the edge AC , the number of edges in a maximum matching is still M . From the choice of G , there exists a vertex cover with M vertices. The vertices in this vertex cover must be the endpoints of the edges in the maximum matching. So B is not one of them. Since the edge AB is covered, A must be a vertex in the vertex cover. It follows that this vertex cover is also a vertex cover of G . But this implies $M = N$, which is a contradiction.

So we must have $M = N$.

41. We use terminologies in graph theory. The boys and girls give a natural bipartition of the graph. Let M be the number of edges in a maximum matching. Suppose on the contrary that $M \leq 11$. By König's theorem, we can choose M vertices such that every edge has an endpoint being one of these vertices. We remove any 11 vertices including these M vertices. From the condition, there is an edge among the remaining 31 vertices, which is a contradiction. So we have $M \geq 12$.